

Dual-Channel Temporal Graph Learning for Real-Time Detection of Coordinated and Previously Unseen Payment Fraud

Md Sibbir Hossain

Independent Researcher, New York, NY, USA

ORCID: 0009-0002-0795-4512

Abstract

Organized payment fraud distributes activity across many cards that share devices, network identities, and cash-out merchants, so individual transactions can appear benign. Supervised detectors additionally fail on attack patterns absent from their training data. We present FinGuard-XAI, an open-source streaming system that (i) maintains a point-in-time heterogeneous temporal graph of cards, devices, merchants, and network identities; (ii) scores each transaction with a temporal GraphSAGE model over its sampled two-hop neighborhood; (iii) adds a *dual-channel* alerting scheme in which a Deep SVDD novelty channel with a fixed, auditable false-positive budget operates alongside the calibrated supervised channel; and (iv) explains each alert with GNNExplainer. In controlled experiments over five independently generated synthetic datasets with chronological splits, the temporal GNN reaches a PR-AUC of 0.933 versus 0.701 for gradient-boosted trees on the same entity features, while lowering the false-positive rate from 2.50% to 0.60%. When account takeover is withheld from training, supervised-only recall on this unseen pattern is 0.059, and the dual-channel design raises it to 0.335. Under simulated camouflage and device-dispersion attacks, the dual-channel design maintains ring recall of at least 0.97, at the cost of roughly doubling the false-positive rate. Per-transaction scoring takes 4.7 ms at the 95th percentile on a single CPU thread. Code, data generator, and all experiment logs are public.

1 Introduction

Fraud losses in the United States are large and rising. Consumers reported losing more than \$12.5 billion to fraud in 2024, a 25% increase over 2023 [1]. The FBI’s Internet Crime Complaint Center recorded \$16.6 billion in reported cybercrime losses in the same year [2], and the U.S. Government Accountability Office estimates federal fraud losses at \$233–\$521 billion per year [3]. Two properties of modern fraud limit conventional detectors. First, organized fraud is *relational*: the evidence lies in shared devices, network identities, and merchants rather than in any single transaction. Second, fraud is *adaptive*: new tactics appear that are absent from labeled training data.

Graph neural networks (GNNs) address the first property [5, 6, 8, 9, 10]. The second is typically addressed by unsupervised anomaly detection [13, 12], and hybrid combinations are common, including in the architecture this work builds on [4]. We observe a practical failure mode of such hybrids. When the fusion of supervised and anomaly scores is learned on validation data that contain no novel fraud, the fusion learns to ignore the anomaly signal, and the benefit disappears exactly where it is needed.

Contributions.

1. An open, streaming implementation of temporal graph fraud detection with point-in-time feature construction, verified by automated leakage tests.
2. A *dual-channel* alerting scheme that separates known-pattern risk from novelty, with the novelty channel governed by an explicit false-positive budget on legitimate traffic.
3. A multi-seed evaluation protocol covering a strong non-graph baseline, an unseen-fraud hold-out, and adaptive camouflage and dispersion attacks, with all numbers reported with uncertainty and all logs released.

2 Related Work

GraphSAGE [5] and GAT [6] are standard inductive GNNs. Fraud-specific GNNs address camouflage and class imbalance [8, 9], and GADBench [10] shows that tree ensembles with graph features are strong baselines, which motivates our GBDT comparison. Temporal Graph Networks [7] introduced functional time encodings, which we adopt. Graph learning for financial forensics was demonstrated on Bitcoin transaction data [11]. Deep SVDD [12] and Isolation Forest [13] are established one-class detectors. The difficulty of recognizing classes absent at training time is studied as open-set recognition [14]. GNNExplainer [15] provides post-hoc subgraph explanations. Our contribution is not a new network architecture but a system design and evaluation that explicitly measure behavior on unseen and adaptive fraud.

3 Method

Temporal graph. Each transaction t with card c , device d , merchant m , network identity i , amount a and time τ inserts undirected typed edges (c, d) , (c, m) , (c, i) , (d, i) with features $[\log(1+a), \Delta\tau, \text{relation}]$. Each node keeps its 64 most recent interactions, which bounds memory and latency. Node features (type, activity count, mean log-amount, number of distinct neighbors, recency, age) are computed from past events only.

Model. For each transaction we extract the two-hop neighborhood of its four entities (8 most recent neighbors per hop). Edge time gaps are encoded as $\phi(\Delta\tau) = \cos(w\Delta\tau + b)$ with learned w, b [7]. Time-encoded edge features are mean-aggregated onto nodes, followed by two GraphSAGE layers (hidden size 64). The transaction representation z_t concatenates the four entity embeddings with transaction features. A two-layer MLP outputs a logit s_t , trained with class-weighted binary cross-entropy.

Dual-channel alerting. A Deep SVDD network ψ is fit on representations of legitimate training transactions, giving the anomaly score $v_t = \|\psi(z_t) - \mathbf{c}\|^2$. Channel 1 is a logistic-regression stacker over $(s_t, \log(1 + v_t))$, fitted on a later validation window, with an F1-optimal threshold θ_1 . Channel 2 raises an alert when $v_t > \theta_2$, where θ_2 is the $(1 - \beta)$ quantile of v on legitimate validation traffic. The budget β (default 0.5%) directly bounds the novelty channel’s false-positive rate on in-distribution legitimate traffic. The final alert is $\mathcal{K}[p_t \geq \theta_1] \vee \mathcal{K}[v_t > \theta_2]$.

Explanations. For each alert, GNNExplainer [15] optimizes node and edge masks on the transaction subgraph. It runs on a separate model copy, outside the scoring path.

4 Experimental Setup

Data. We use a documented synthetic generator (no personal data) with 20,000 legitimate transactions per seed from 3,000 cards over 30 days. Legitimate users occasionally appear on new devices and networks. Fraud rings of 4–9 cards route activity through 1–3 *already-active* (compromised) devices and network identities toward 1–2 colluding merchants over 48 hours. Account takeover (ATO) uses a new device and a common network. Fraud amounts overlap with legitimate amounts. A loader for the public IEEE-CIS dataset is included; results on it are future work.

Protocol. Chronological 70/15/15 split with thresholds chosen on validation only; 5 seeds, each with an independently generated dataset; mean $\pm$ standard deviation (tables) and 95% bootstrap intervals (released). Baselines: HistGradientBoosting on identical entity and transaction features *without* message passing; Deep SVDD alone; the supervised GNN alone. The *unseen-fraud* setting removes all ATO from training and validation. The *robustness* setting freezes the trained system and perturbs only future ring activity with (a) device dispersion (fresh device and network per transaction), (b) camouflage (two benign-looking purchases at popular merchants from the card’s usual device before each fraudulent one), or (c) both.

5 Results

Table 1: Standard benchmark (all fraud types seen in training); 5 seeds.

Model	PR-AUC	Recall	Precision	FPR
GBDT (no graph)	0.701 ± 0.128	0.719 ± 0.100	0.563 ± 0.263	2.50%
Deep SVDD only	0.677 ± 0.159	0.711 ± 0.083	0.609 ± 0.247	1.88%
Temporal GNN	0.933 ± 0.032	0.895 ± 0.036	0.822 ± 0.051	0.60%
GNN + dual-channel	0.816 ± 0.086	0.912 ± 0.038	0.714 ± 0.061	1.14%

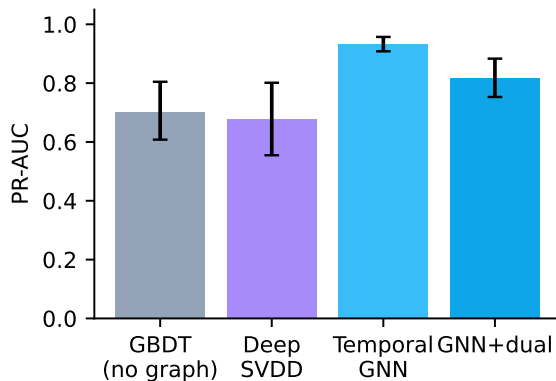


Figure 1: PR-AUC with 95% bootstrap intervals across 5 seeds.

Relational signal. The temporal GNN substantially outperforms the non-graph GBDT on the same entity features (Table 1): higher PR-AUC and recall at roughly one quarter of the false-

positive rate. In the standard setting the dual-channel design trades some precision and PR-AUC for higher recall, because the novelty channel re-ranks borderline transactions.

Table 2: Unseen-fraud setting: ATO removed from training and validation; 5 seeds.

Model	PR-AUC	Recall	Precision	FPR
GBDT (no graph)	0.439 ± 0.185	0.357 ± 0.145	0.589 ± 0.251	1.00%
Deep SVDD only	0.640 ± 0.182	0.651 ± 0.135	0.541 ± 0.198	2.02%
Temporal GNN	0.757 ± 0.083	0.585 ± 0.128	0.913 ± 0.061	0.19%
GNN + dual-channel	0.733 ± 0.095	0.705 ± 0.088	0.724 ± 0.036	0.91%

Unseen fraud. Without ATO in training, the supervised GNN detects almost none of it (recall 0.059 ± 0.067). The stacker assigns near-zero weight to the anomaly score, because validation data contain no novel fraud. The novelty channel raises ATO recall to 0.335 ± 0.068 while ring recall is unchanged, and overall FPR rises from 0.19% to 0.91% (Table 2, Fig. 2a).

Table 3: Fraud-ring recall under adaptive attacks applied after training; 5 seeds.

Attack	Recall (GNN)	Recall (dual)	FPR (GNN)	FPR (dual)
none	0.973 ± 0.037	0.973 ± 0.037	0.60%	1.28%
device dispersion	0.981 ± 0.026	0.988 ± 0.018	0.59%	1.36%
camouflage	0.918 ± 0.043	0.968 ± 0.037	0.66%	1.43%
combined	0.839 ± 0.201	0.976 ± 0.030	0.66%	1.44%

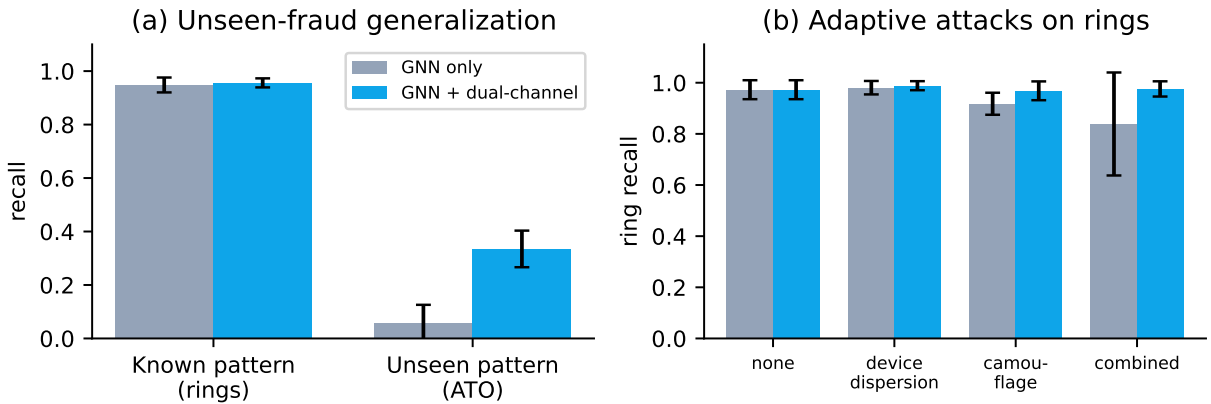


Figure 2: (a) Recall on known vs. unseen fraud. (b) Ring recall under adaptive attacks. Error bars: std over 5 seeds.

Adaptive attacks. Device dispersion alone did not reduce detection. Brand-new devices are themselves anomalous, and rings still converge on colluding merchants. Camouflage lowered supervised recall, and the combined attack made it unstable across seeds (Table 3). The dual-channel design kept ring recall close to its no-attack level in all settings, at approximately double the false-positive rate.

Latency. Averaged over seeds, per-transaction scoring (graph update, subgraph extraction, GNN, Deep SVDD, fusion) took 3.8/4.7/5.5 ms at p50/p95/p99 on one CPU thread. Explanations took about 100 ms (median), computed only for alerts and off the scoring path.

6 Limitations

All headline results use synthetic data whose generator encodes our assumptions; absolute numbers will differ on real data, and evaluation on public benchmarks (IEEE-CIS) is the next step. Ring counts per test window are small (32–125 transactions), so robustness estimates carry high variance. The attacks are heuristic, and white-box adaptive attackers are not evaluated. The novelty channel’s FPR budget holds for in-distribution legitimate traffic and may be exceeded under benign distribution shift; we therefore expose a PSI drift monitor. GNNExplainer outputs describe model behavior, not causation. The prototype uses a single-process in-memory graph.

7 Conclusion

Relational, time-aware learning substantially improves the detection of coordinated fraud over a strong non-graph baseline in our controlled setting. Learned fusion of supervised and anomaly scores can silently discard the anomaly signal. A separately budgeted novelty channel recovers part of the detection of unseen fraud and stabilizes detection under camouflage attacks, at an explicit and tunable false-positive cost. Code and logs: <https://github.com/sibbirhossain/finguard-xai>.

References

- [1] Federal Trade Commission. New FTC data show a big jump in reported losses to fraud to \$12.5 billion in 2024. Press release, March 2025.
- [2] Federal Bureau of Investigation, Internet Crime Complaint Center. *Internet Crime Report 2024*. 2025.
- [3] U.S. Government Accountability Office. Fraud Risk Management: 2018–2022 Data Show Federal Government Loses an Estimated \$233 Billion to \$521 Billion Annually to Fraud. GAO-24-105833, April 2024.
- [4] R. M. Zakaria, M. M. Rahman, M. T. H. Choudhury, H. Rahman, M. A. Rafi, A. Minto, M. S. Hossain, and S. I. Saimon. Detecting financial fraud in real-time transactions using graph neural networks and anomaly detection techniques. *Journal of Economics, Finance and Accounting Studies*, 7(6):1–13, 2025.
- [5] W. Hamilton, Z. Ying, and J. Leskovec. Inductive representation learning on large graphs. In *NeurIPS*, 2017.
- [6] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio. Graph attention networks. In *ICLR*, 2018.
- [7] E. Rossi, B. Chamberlain, F. Frasca, D. Eynard, F. Monti, and M. Bronstein. Temporal graph networks for deep learning on dynamic graphs. arXiv:2006.10637, 2020.
- [8] Y. Dou, Z. Liu, L. Sun, Y. Deng, H. Peng, and P. S. Yu. Enhancing graph neural network-based fraud detectors against camouflaged fraudsters. In *CIKM*, 2020.
- [9] Y. Liu, X. Ao, Z. Qin, J. Chi, J. Feng, H. Yang, and Q. He. Pick and choose: A GNN-based imbalanced learning approach for fraud detection. In *The Web Conference*, 2021.

- [10] J. Tang, F. Hua, Z. Gao, P. Zhao, and J. Li. GADBench: Revisiting and benchmarking supervised graph anomaly detection. In *NeurIPS Datasets and Benchmarks*, 2023.
- [11] M. Weber, G. Domeniconi, J. Chen, D. K. I. Weidele, C. Bellei, T. Robinson, and C. E. Leiserson. Anti-money laundering in Bitcoin: Experimenting with graph convolutional networks for financial forensics. KDD Workshop on Anomaly Detection in Finance, 2019.
- [12] L. Ruff, R. Vandermeulen, N. Görnitz, L. Deecke, S. A. Siddiqui, A. Binder, E. Müller, and M. Kloft. Deep one-class classification. In *ICML*, 2018.
- [13] F. T. Liu, K. M. Ting, and Z.-H. Zhou. Isolation forest. In *ICDM*, 2008.
- [14] W. J. Scheirer, A. de Rezende Rocha, A. Sapkota, and T. E. Boult. Toward open set recognition. *IEEE TPAMI*, 35(7):1757–1772, 2013.
- [15] R. Ying, D. Bourgeois, J. You, M. Zitnik, and J. Leskovec. GNNExplainer: Generating explanations for graph neural networks. In *NeurIPS*, 2019.